STATS 210 **Task 1** **Some Answers**

(a) The key is to bring out that the 40% is selecting from the **Irish customers**. Some possible sentences:

(i) 40% of the Irish customers play the banjo.

(ii) Of the Irish customers, 40% play the banjo.

(iii) The probability that an Irish customer plays the banjo is 0.4.

(iv) The probability of playing the banjo is 0.4 for the Irish customers.

(v) Out of the Irish customers, the proportion of banjo-players is 40%.

(vi) Given that a customer is Irish, the probability he or she plays the banjo is 0.4.

(b) The key is to bring out that the 20% is selecting from **all customers**. Some possible sentences:

(i) The probability that a customer is an Irish banjo-player is 0.2.

(ii) 20% of the customers are Irish and play the banjo.

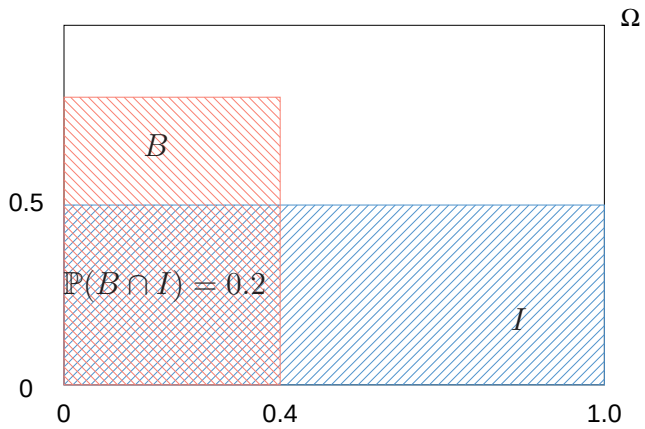
(iii) A customer is Irish and plays the banjo with probability 0.2.

(c) It's easiest to convey areas precisely if you use rectangles instead of circles on the Venn diagram. The diagram below shows the intersection area as being exactly fraction 0.2 of the whole, and exactly fraction 0.4 of the blue area (event I). A scale is marked on the diagram.

Note that

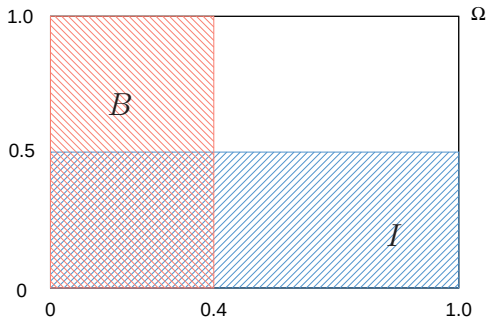
$$0.4 = \mathbb{P}(B \mid I) = \frac{\mathbb{P}(B \cap I)}{\mathbb{P}(I)} = \frac{0.2}{\mathbb{P}(I)} \quad \Rightarrow \quad \mathbb{P}(I) = 0.5.$$

However, we cannot calculate $\mathbb{P}(B)$ from the information given, so B can be any height in the diagram below.

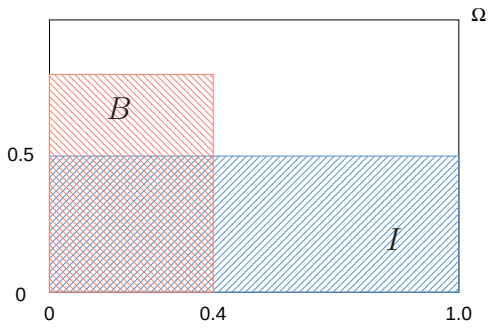


$\mathbb{P}(B \mid I) = 0.4$

(d)(i) If B and I are independent, then $\mathbb{P}(B) = \mathbb{P}(B \mid I) = 0.4$, and $\mathbb{P}(I \mid B) = \mathbb{P}(I) = 0.5$.



(ii) Any other diagram (like in part (c)) will show the events being non-independent:



(e) If B and I are independent, they occur in the same proportions regardless of whether they happen together or separately. I occurs half the time overall, so it will also occur on half the times that B happens, so $P(I \cap B) = 0.5\mathbb{P}(B) = \mathbb{P}(I)\mathbb{P}(B)$.

Wording from previous teams: The following are examples of excellent answers.

The probability that a customer is Irish is the same regardless of whether they play the banjo or not, and vice versa. Therefore since 50% of customers are Irish and 40% play the banjo it makes sense to say that 50% of the 40% that play the banjo are Irish, or $0.5 \times 0.4 = 0.2$.

Statistically independent events exert no influence over one another. If event A happens 50% of the time and B 50%, then both events occurring will involve A happening 50% of the time that B occurs, which is 25% of the total time.

Say I happens 50% of the time. B doesn't affect I, so I happens 50% of the time B happens, therefore $\mathbb{P}(I \cap B) = 0.5\mathbb{P}(B) = \mathbb{P}(I)\mathbb{P}(B)$.

If B and I are independent then the fraction of the sample space taken up by B is equal to the fraction of I taken up by B so the probability of the overlap equals $\mathbb{P}(B)\mathbb{P}(I)$.